

SE3110 — Software Testing & Quality Evaluation

Comprehensive Test Case Specification Document

Target System: WayPoint — Integrated Full-Stack & Agentic AI Transit Platform

📌 Document Control & Metadata

Field	Detail
Module	SE3110: Quality Management in Software Engineering
Academic Year / Semester	Year 3, Semester 1 — Academic Year 2026
Institution	Sri Lanka Institute of Information Technology (SLIIT)
Project Title	WayPoint Transit Operations & Journey Planning Platform
Document Type	Comprehensive Test Case Specification Repository
Version	1.0 (Final Release)
Date	8th October 2026
Total Test Cases	68 Structured Test Specifications (Traceable to 448 Automated Executions)
Overall Pass Rate	100% (448 / 448 automated test instances passed)

👥 Student Component & Responsibility Matrix

Student Name	Student IT Number	Assigned Component	Testing Focus & Scope
Vinranga M.W.S	IT24100374	Component 1: Journey Planning & Routes	Route catalogue, timetables, connecting transfer window logic (BR-TRANSFER-001), journey search, Journey Agent.
Mohomed N.M.N. (Leader)	IT24102476	Component 2: Fleet, 2D Seats & Operations	Bus fleet inventory, 2D seat map designer, driver rostering rest hours (BR-TIME-001), Resource Feasibility Agent.
Dissanayaka A.D.M.N.K.	IT24100225	Component 3: Seat Hold, Payments & QR	10-minute hold concurrency lock (BR-HOLD-001), payment sandbox, HMAC-SHA256 QR tickets, tiered refunds (BR-REFUND-001).
Dineth sasmitha J.S.D.	IT24102912	Component 4: Disruption, Approval & Safety	Disruption intake, passenger blast radius, Transport Manager approval gate (BR-APPROVAL-001), conductor scanner, Safety Agent.

1. Testing Category 1: Authoritative Backend & API Testing (.NET 8 / xUnit)

Test Case ID	Component / Feature	Test Type	Preconditions	Inputs & Execution Steps	Expected Result	Actual Result	Status	Traceability
TC-BE-001	Booking / Tiered Refund	Unit (Normal)	Booking exists \$>24\$h before departure	Calculate refund for Rs. 5,700 fare at \$T-30\$ hours.	90% refund (Rs. 5,130), 10% cancellation fee retained.	Exact Rs. 5,130 refund calculated.	Passed	BR-REFUND-001
TC-BE-002	Booking / Tiered Refund	Unit (Boundary)	Booking exists \$12\text{-}24\$h before departure	Calculate refund for Rs. 5,700 fare at \$T-18\$ hours.	50% refund (Rs. 2,850), 50% cancellation fee.	Exact Rs. 2,850 refund calculated.	Passed	BR-REFUND-001

Test Case ID	Component / Feature	Test Type	Preconditions	Inputs & Execution Steps	Expected Result	Actual Result	Status	Traceability
TC-BE-003	Booking / Tiered Refund	Unit (Edge/Fail)	Booking exists \$ < 12\$h before departure	Calculate refund for Rs. 5,700 fare at \$T-6\$ hours.	0% refund (Rs. 0.00), non-refundable message returned.	Rs. 0 refund returned; cancellation fee 100%.	Passed	BR-REFUND-001
TC-BE-004	Security / HMAC QR Ticket	Unit (Normal)	Legitimate booking reference WP-7B92K1	Invoke GenerateTicketPayloadAsync with valid booking data.	Returns HMAC-SHA256 signed payload string with WP prefix.	Valid 64-char HMAC token produced and verified.	Passed	BR-HMAC-001
TC-BE-005	Security / HMAC QR Ticket	Unit (Invalid/Tamper)	Authentic QR ticket generated	Modify seat from 4A to 1A in payload and call VerifyTicketQrAsync.	Verification fails with Security Violation: Tampered Signature.	IsValid=False; Security violation flagged.	Passed	BR-HMAC-001
TC-BE-006	Booking / Payment Sandbox	Unit (Normal)	Sandbox gateway active	Submit card 4000 0000 0000 0001 with amount Rs. 5,700.	Gateway returns IsSuccess=True with TXN- transaction ID.	Status Success, TXN- generated.	Passed	US-PASS-004
TC-BE-007	Booking / Payment Sandbox	Unit (Failure)	Sandbox gateway active	Submit card 4000 0000 0000 0002 (declined preset).	Gateway returns IsSuccess=False, GatewayStatus= Declined.	IsSuccess=False, Declined status returned.	Passed	US-PASS-004
TC-BE-008	Disruption / Impact Severity	Unit (Normal)	Disrupted service logged	Timetable shift is 10 minutes (\$ < 15\$ min).	Impact classified as Low; auto-executable by operator.	Impact= Low returned.	Passed	BR-DISRUPT-001
TC-BE-009	Disruption / Impact Severity	Unit (Boundary)	Disrupted service logged	Timetable shift is exactly 15 minutes.	Impact classified as Low (inclusive threshold).	Impact= Low returned.	Passed	BR-DISRUPT-001
TC-BE-010	Disruption / Impact Severity	Unit (Boundary)	Disrupted service logged	Timetable shift is 16 minutes (\$ > 15\$ min).	Impact classified as High; requires Transport Manager sign-off.	Impact= High returned.	Passed	BR-APPROVAL-001
TC-BE-011	Fleet / Driver Rest Rules	Unit (Normal)	Driver assigned to Shift A ending 10:00	Assign Driver to Shift B starting 19:00 (9 hrs gap).	Assignment accepted; rest buffer exceeds mandatory 8 hours.	Validation passed; no overlap.	Passed	BR-TIME-001
TC-BE-012	Fleet / Driver Rest Rules	Unit (Invalid)	Driver assigned to Shift A ending 10:00	Assign Driver to Shift B starting 15:00 (5 hrs gap).	Assignment rejected; violates 8-hour mandatory rest gap.	Validation failed with labor safety exception.	Passed	BR-TIME-001

2. Testing Category 2: Relational Database & Concurrency Testing (PostgreSQL / EF Core)

Test Case ID	Component / Feature	Test Type	Preconditions	Inputs & Execution Steps	Expected Result	Actual Result	Status	Traceability	Member
TC-DB-001	SeatHold / Concurrency Token	Integration (Race Condition)	Service active, seat 12A available in DB	Fire two parallel asynchronous hold requests for seat 12A simultaneously.	Exactly 1 request receives HTTP 200/201 OK; exactly 1 request receives HTTP 409 Conflict.	Request 1 = 200 OK; Request 2 = 409 Conflict. Zero double-holds.	Passed	REQ-TEST-02	Mithila
TC-DB-002	SeatHold / Expiry Lifecycle	Integration (Boundary)	Hold created at \$T_0\$ with 10-minute expiry	Query seat status at \$T_0 + 9\text{(m)}50\text{(s)}\$ vs \$T_0 + 10\text{(m)}01\text{(s)}\$.	Seat status transitions from Held to Available after 10 minutes.	Hold expired automatically; seat returned to available inventory.	Passed	BR-HOLD-001	Mithila

Test Case ID	Component / Feature	Test Type	Preconditions	Inputs & Execution Steps	Expected Result	Actual Result	Status	Traceability	Member
TC-DB-003	Identity / Unique Constraint	Integration (Invalid)	User <code>passenger@waypoint.lk</code> exists in DB	Attempt <code>INSERT</code> with duplicate email address.	PostgreSQL throws 23505 <code>UniqueViolation</code> ; EF Core throws <code>DbUpdateException</code> .	HTTP 400/409 duplicate registration rejected.	Passed	Security / DB	Nuhadh
TC-DB-004	Audit / Log Hash Chain	Integration (Data Integrity)	Operational change committed	Verify SHA-256 tamper-evident hash chain linking log entries.	Each audit entry hashes previous record's hash + current payload.	Hash integrity verified across log series.	Passed	REQ-AUDIT-01	Dineth
TC-DB-005	Fleet / Seat Layout Matrix	Integration (Constraint)	Bus layout defined with 10 rows $\times$ 4 cols	Insert seats violating foreign key or matrix bounds.	Constraint violation prevents orphaned seat records.	Relational integrity strictly maintained.	Passed	REQ-FLEET-01	Nuhadh

3. Testing Category 3: React Web Frontend Testing (Vitest & RTL)

Test Case ID	Component / Feature	Test Type	Preconditions	Inputs & Execution Steps	Expected Result	Actual Result	Status	Traceability	Member
TC-WEB-001	Journey / RouteManager	Component (Normal)	Operator logged in	Render <code>RouteManagerPage</code> with mocked catalogue data.	Renders route list table, corridor badges, and stop creation modal.	Component rendered cleanly; all columns visible.	Passed	REQ-TEST-03	Sethum
TC-WEB-002	Fleet / SeatLayoutDesigner	Component (Normal)	Fleet manager logged in	Render <code>SeatLayoutDesignerPage</code> and select 2+2 layout preset.	Generates 4-column seat grid with central aisle gap visualization.	Grid rendered; seat click toggles availability state.	Passed	REQ-TEST-03	Nuhadh
TC-WEB-003	Bookings / ManifestMonitor	Component (Normal)	Active service selected	Render <code>BookingManifestMonitorPage</code> with passenger list.	Displays real-time manifest, hold countdown monitor, and search bar.	Filter input filters passenger rows instantly.	Passed	REQ-TEST-03	Mithila
TC-WEB-004	Disruptions / DisruptionHub	Component (Normal)	Transport Manager logged in	Render <code>DisruptionHubPage</code> with high-impact rebooking proposal.	Disruption details shown; Manager Approval button enabled.	"Approve Rebooking" triggers action handler.	Passed	REQ-TEST-03	Dineth
TC-WEB-005	Admin / UserGovernance	Component (Normal)	Admin logged in	Render <code>AdminUsersPage</code> with locked user account.	Displays lockout badge and provides "Unlock Account" button.	Account unlocked; table state reflects active status.	Passed	REQ-SEC-01	Nuhadh

Test Case ID	Component / Feature	Test Type	Preconditions	Inputs & Execution Steps	Expected Result	Actual Result	Status	Traceability	Member
TC-WEB-006	Auth / RBAC Route Guard	Protected Route	Unauthenticated visitor	Attempt navigation to <code>/admin/users</code> or <code>/disruptions/manage</code> .	Protected route intercepts navigation and redirects to <code>/login</code> .	User redirected to <code>/login</code> with <code>returnUrl</code> set.	Passed	REQ-SEC-02	Sethum

4. Testing Category 4: Flutter Mobile Client Testing (flutter_test)

Test Case ID	Component / Feature	Test Type	Preconditions	Inputs & Execution Steps	Expected Result	Actual Result	Status	Traceability	Member
TC-MOB-001	Journey / SearchScreen	Widget (Normal)	App launched on Explore tab	Select <code>Origin="Colombo"</code> , <code>Dest="Kandy"</code> , <code>Date="2026-10-15"</code> .	Displays timetable card list with departure times and fare badges.	Journey cards displayed with corridor chips.	Passed	REQ-TEST-04	Sethum
TC-MOB-002	Fleet / SeatPickerBloc	Unit (State Transition)	Service seat layout loaded	Dispatch <code>SelectSeat("12A")</code> followed by <code>HoldSeats()</code> .	State transitions: <code>Loaded</code> $\rightarrow$ <code>SeatsHolding</code> $\rightarrow$ <code>SeatsHeld</code> .	State emitted with 600s countdown timer.	Passed	REQ-TEST-04	Nuhadh
TC-MOB-003	Fleet / SeatPickerBloc	Unit (Invalid/Duplicate)	Seat <code>12A</code> already selected	Dispatch <code>SelectSeat("12A")</code> again.	Duplicate selection ignored; selection count remains 1.	Duplicate rejected; state preserved.	Passed	REQ-TEST-04	Nuhadh
TC-MOB-004	Fleet / ReviewSubmission	Widget (Validation)	Passenger completed journey	Tap "Submit Review" with 0 stars selected.	Form displays error: "Please select a star rating".	Validation banner displayed; API call blocked.	Passed	REQ-TEST-04	Nuhadh
TC-MOB-005	Tickets / TicketWallet	Widget (Normal)	Confirmed booking exists	Navigate to Tickets tab.	Renders dynamic QR barcode and brightness booster control.	QR widget rendered with booking reference <code>WP-XXXXXX</code> .	Passed	REQ-TEST-04	Mithila
TC-MOB-006	Disruptions / AlertBanner	Widget (Normal)	Disruption broadcasted	Push alert received for booked service.	Displays persistent alert banner with delay info and reroute choice.	Alert card displayed with action buttons.	Passed	REQ-TEST-04	Dineth
TC-MOB-007	Conductor / QRScanner	Widget (Normal)	Conductor logged in	Simulate camera scan of valid QR ticket payload.	Displays green validation checkmark and passenger boarding button.	Valid ticket confirmed; boarding status updated.	Passed	REQ-TEST-04	Dineth

5. Testing Category 5: Closed-Loop Cross-Platform Integration & E2E Testing

Test Case ID	Workflow Step	Test Level	Triggering Actor	Execution Steps & Target Endpoints	Expected System Result	Actual Result	Status	Traceability	Member
TC-E2E-001	Step 1: Health & Connectivity	E2E Integration	System Client	GET /health	Status=200, database="Connected".	HTTP 200, DB healthy.	Passed	Baseline	Shared
TC-E2E-002	Step 2: Journey Exploration	E2E Integration	Passenger (Mobile)	POST /api/v1/journeysearch/search	Returns matching bus services between Colombo & Kandy.	Search returned valid service payload.	Passed	BR-TRANSFER-001	Sethum
TC-E2E-003	Step 3: Interactive Seat Hold	E2E Integration	Passenger (Mobile)	POST /api/v1/seathold (Seat 12A)	Returns 201 Created with 10-minute hold lock; seat unavailable to others.	Hold lock created; HTTP 200/201.	Passed	BR-HOLD-001	Mithila
TC-E2E-004	Step 4: Disruption & AI Proposal	E2E Integration	Dispatcher (Web)	POST /api/v1/disruption/cases	LangGraph AI generates rebooking plan; marks status PendingApproval .	Case logged; impact evaluated.	Passed	FR-AI-003	Dineth
TC-E2E-005	Step 5: Manager Approval Gate	E2E Integration	Manager (Web)	POST /api/v1/approval/decide	Decision recorded; high-impact rebooking plan approved for execution.	Manager approval registered.	Passed	BR-APPROVAL-001	Dineth
TC-E2E-006	Step 6: Checkout & QR Issuance	E2E Integration	Passenger (Mobile)	POST /api/v1/payment/charge	Commits payment, assigns seat, generates HMAC-SHA256 signed QR pass.	QR generated with tamper-proof HMAC.	Passed	BR-HMAC-001	Mithila
TC-E2E-007	Step 7: Conductor Boarding Scan	E2E Integration	Conductor (Mobile)	POST /api/v1/ticket/verify-qr	Validates HMAC signature; marks ticket status as Boarded .	Signature verified; boarding complete.	Passed	Closed-Loop	Dineth

6. Testing Category 6: Non-Functional Testing (Performance & Security — REQUIRED)

Test Case ID	Non-Functional Area	Tool / Framework	Test Configuration	Execution Steps & Payload	Expected Result	Actual Measured Result	Status	Member
TC-NF-PERF-001	Concurrency / Load Testing	k6 & Python Benchmark	20 concurrent VUs, 50 requests	GET /health under concurrent load.	Throughput ≥ 5 req/s, P95 < 5000ms.	7.76 req/s, P50=1243ms, P95=4645ms, 0% errors.	Passed	Mithila
TC-NF-PERF-002	High-Traffic Search Latency	k6 & Python Benchmark	20 concurrent VUs, 50 requests	POST /journeysearch/search concurrent search queries.	P95 latency < 1000ms, 0 server crashes.	28.19 req/s, P50=574ms, P95=624ms, 0% 500 errors.	Passed	Sethum
TC-NF-PERF-003	Seat Hold Collision Contention	k6 & Python Benchmark	25 concurrent VUs, 50 requests	Rapid concurrent seat hold attempts on single seat 12A .	No double bookings; atomic isolation maintained.	12.35 req/s, P50=1924ms, atomic barrier verified.	Passed	Mithila
TC-NF-SEC-001	Forged JWT Token Rejection	Python Requests & Pytest	Forged HS256 JWT token	GET /api/v1/user/profile with tampered role claim (Admin).	Rejects with HTTP 401 Unauthorized; invalid signature.	HTTP 401; access denied.	Passed	Nuhadh

Test Case ID	Non-Functional Area	Tool / Framework	Test Configuration	Execution Steps & Payload	Expected Result	Actual Measured Result	Status	Member
TC-NF-SEC-002	RBAC Privilege Escalation	Python Requests & Pytest	Valid Passenger token	<code>POST /api/v1/bus</code> (attempting fleet vehicle creation).	Rejects with HTTP 403 Forbidden; role unauthorized.	HTTP 403 / 401 ; blocked.	Passed	Nuhadh
TC-NF-SEC-003	SQL Injection Immunity	Python Requests & Pytest	Parameterized EF Core queries	Search query with <code>' OR 1=1 --</code> and <code>'; DROP TABLE;</code> .	No SQL syntax errors; no database schema leaks; status <code>\$\ne 500\$</code> .	Status 200/404 ; 0 crashes; safe parameterized query.	Passed	Sethum
TC-NF-SEC-004	HMAC QR Tamper Detection	Python Pytest & HMAC	Altered seat data in QR string	Modify seat from <code>4B</code> to <code>1A</code> in cryptographic payload.	Signature mismatch detected; rejected immediately.	HMAC signature mismatch flagged ; forgery prevented.	Passed	Mithila
TC-NF-SEC-005	AI Prompt Injection Defusing	Python Pytest & Regex	Adversarial jailbreak string	<code>sanitize_user_input("IGNORE PREVIOUS INSTRUCTIONS...")</code> .	Injection markers stripped; user input safely wrapped in tags.	Injection pattern neutralized ; prompt defused.	Passed	Dineth
TC-NF-SEC-006	Safe-Failure Fault Tolerance	Python Pytest & Asyncio	Simulated 3 cascading outages	Agent fails 3 consecutive times during rebooking execution.	Circuit breaker trips; workflow enters <code>SafeFailure</code> (FR-AI-004).	SafeFailure status emitted ; zero unhandled crashes.	Passed	Dineth

7. Testing Category 7: Agentic AI Testing & Evaluation (Pytest / LangGraph)

Test Case ID	AI Subsystem Area	Test Type	Preconditions	Inputs & Execution Steps	Expected Result	Actual Result	Status	Traceability	Member
TC-AI-001	Tool Registry Strict Allow-List	Evaluation (Security)	Tool registry initialized	Query allow-listed tools in AI microservice.	Exactly 10 registered tools; unauthorized tools rejected.	Exactly 10 tools; strict registry enforced.	Passed	BR-AITool-001	Dineth
TC-AI-002	LangGraph State Transition	Workflow (Normal)	Disruption workflow invoked	Pass valid disrupted service state through LangGraph graph.	5 nodes execute in sequence: Journey $\rightarrow$ Resource $\rightarrow$ Booking $\rightarrow$ Safety $\rightarrow$ End.	All 5 nodes executed successfully.	Passed	REQ-AI-001	Dineth
TC-AI-003	Driver Rest Hours Guardrail	Guardrail (Labor Safety)	Replacement driver assigned	Replacement driver has 6 hours rest prior to shift.	Deterministic guardrail rejects assignment; requires >8 hours.	Guardrail flagged labor violation.	Passed	BR-TIME-001	Nuhadh
TC-AI-004	Bus Seating Capacity Guardrail	Guardrail (Physical Limits)	Replacement vehicle assigned	Replacement bus capacity (40 seats) $<$ affected passengers (45).	Guardrail rejects vehicle; capacity deficit flagged.	Guardrail rejected allocation.	Passed	BR-FLEET-001	Nuhadh
TC-AI-005	Manager Approval Gate Override	Guardrail (Governance)	AI classifies disruption as minor	Route cancellation occurs or delay exceeds 15 minutes.	Deterministic guardrail forces classification to <code>HighImpact</code> .	Guardrail overrode to HighImpact; Manager Gate engaged.	Passed	BR-APPROVAL-001	Dineth

Test Case ID	AI Subsystem Area	Test Type	Preconditions	Inputs & Execution Steps	Expected Result	Actual Result	Status	Traceability	Member
TC-AI-006	Safe-Failure Consecutive Trip	Guardrail (Resilience)	Agent encounters network error	Re-attempt execution up to retry threshold (\$N=3\$).	Workflow marks status as SafeFailure; zero silent failures.	SafeFailure triggered on 3rd failure.	Passed	FR-AI-004	Dineth

8. Test Execution Summary Matrix

WAYPOINT AUTOMATED TEST EXECUTION SUMMARY				
Test Suite / Subsystem	Tool / Framework	Total Tests	Passed	Failed
1. Authoritative Backend (.NET 8)	xUnit / Moq / Fluent	82	82	0
2. Web Frontend Single-Page App	Vitest / Testing Library	36	36	0
3. Flutter Mobile Application	flutter_test / mocktail	65	65	0
4. Agentic AI Subsystem (Python)	Pytest / LangGraph	246	246	0
5. Closed-Loop E2E Workflow	Python / Requests / Pytest	7	7	0
6. Cross-Service API & DB Integration	Python / Requests / Pytest	6	6	0
7. Automated Security Audit & Pen-Test	Python / Pytest / Cryptography	6	6	0
TOTAL AUTOMATED TEST VERIFICATIONS		448	448	0 (100% PASS)